Suggested Answers and Commentary

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1. (a) The alternation concerns the morphological marking of subjects: subjects marked with *-ga* can alternatively be marked with *-no*.

   (b) The distribution of the *-ga/-no* alternation in the given examples can be summarized as follows:

Syntactic environment	Example(s)	Availability
Relativization of a direct object	(1), (6)	Yes
Relativization of an indirect object	(7)	No
Long-distance relativization of a direct object (matrix subject)	(9)	Yes
Long-distance relativization of a direct object (embedded subject)	(9)	No
Comparative clause	(3)	Yes
-koto complement clause	(2)	Yes
-to complement clause	(4)	No
Independent declarative clause	(5)	No

   Some possible empirical generalizations are the following:

   i. The *-ga/-no* alternation is optional: wherever *-no* is available, *-ga* is also available.
   ii. The alternation can apply to definite subjects.
   iii. The alternation is available in relative clauses and comparative clauses.
   iv. The alternation is available in *-koto* complement clauses, but not in *-to* complement clauses or independent declarative clauses.
   v. In long-distance relativization of an object, the *-ga/-no* alternation is not available on the subject of the most deeply embedded clause involved in the relativization. However, it can occur on the matrix subject.
   vi. No *-o/-no* conversion: The marker *-o* on a direct object cannot alternate with *-no* (cf., (8)).

   These are examples of empirical generalizations supported by the given observations, rather than an exhaustive characterization of the distribution of the alternation.

   (c) The availability of the *-ga/-no* alternation can depend on the grammatical relation of the displaced *wh*-phrase. In particular, in the given constructions, the alternation is available when a direct object undergoes *wh*-displacement,

but not when an indirect object undergoes *wh*-displacement while a direct object remains in the same clause.

2. (a) The alternation concerns the relative order of the subject and the finite verb: the ordinary subject–verb order alternates with a verb–subject order. The latter is known as *stylistic inversion*.

   (b) The distribution of stylistic inversion in the given examples can be summarized as follows:

Syntactic environment	Example(s)	Availability
Relativization of a prepositional complement	(10)	Yes
Relativization of a direct object	(17)	Yes
Relativization of an indirect object	(18)	No
Long-distance object *wh*-question (embedded clause)	(21)	Yes
Direct-object *wh*-question	(20)	Yes
Temporal-adjunct *wh*-question (without a direct object)	(13)	Yes
Temporal-adjunct *wh*-question (with a direct object)	(19)	No
Comparative clause	(11), (12)	Yes
Subjunctive complement clause	(14)	Yes
Indicative complement clause	(15)	No
Independent declarative clause	(16)	No

   Some possible empirical generalizations are the following:

   i. Stylistic inversion is optional: wherever inversion is available, the ordinary subject–verb order is also available.
   ii. The alternation can apply to definite subjects.
   iii. Stylistic inversion is available in relative clauses, comparative clauses, and certain *wh*-questions.
   iv. Stylistic inversion is available in subjunctive complement clauses, but not in indicative complement clauses or independent declarative clauses.
   v. In long-distance *wh*-displacement of an object, stylistic inversion is available in the most deeply embedded clause involved in the displacement.
   vi. No object inversion: Stylistic inversion involves the subject, not a direct or indirect object.

   These are examples of empirical generalizations supported by the given observations, rather than an exhaustive characterization of the distribution of stylistic inversion.

   (c) The availability of stylistic inversion can depend on the grammatical relation of the displaced *wh*-phrase. In particular, in the given constructions, stylistic inversion is available when a direct object undergoes *wh*-displacement, but not when an indirect object or a temporal adjunct undergoes *wh*-displacement while a direct object remains in the same clause.

3. (a) Island sensitivity can serve as an empirical diagnostic for displacement.

(b) Reconstruction effects, weak crossover effects, and the licensing of parasitic gaps are among the phenomena that can be used to diagnose *wh*-displacement.

- Weak crossover effects arise when a particular interpretation involving a pronoun is unavailable in configurations such as *Who*$_i$ did his$_i$ mother see t_i?, where the displaced phrase crosses a phrase containing a pronoun intended to be coreferential with it.
- Consider *John filed the article without reading it*, where *the article* and *it* are coreferential. The corresponding sentence without *it*, *John filed the article without reading*, is unacceptable. However, a corresponding gap is licensed in *I wonder which article John filed without reading*. A gap licensed in this way, in association with another dependency, is called a *parasitic gap*.

These diagnostics would need to be applied to the relevant dependencies in order to investigate whether they involve *wh*-displacement.

(c) Comparative clauses, for example, may also be analyzed in terms of *wh*-movement. A classical discussion of such an analysis can be found in Chomsky (1977).

- Chomsky, Noam (1977). On *Wh*-Movement. In Peter Culicover, Thomas Wasow, and Adrian Akmajian (eds.), *Formal Syntax*, 71–132. New York: Academic Press.

(d) The Japanese *-ga/-no* alternation and French stylistic inversion exhibit several striking parallels.

i. Both alternations are optional. In environments where the alternative form is available, the form without the alternation is also acceptable.

ii. Both alternations can apply to definite subjects. Moreover, both concern subjects, rather than direct or indirect objects.

iii. Both alternations can occur in relative clauses and comparative clauses. Neither alternation is available in the independent declarative clauses considered here.

iv. In Japanese, the *-ga/-no* alternation is available in *-koto* complement clauses, but not in *-to* complement clauses. In French, stylistic inversion is available in subjunctive complement clauses, but not in indicative complement clauses. This parallel provides a motivation for comparing Japanese *-koto* complements with French subjunctive complements.

v. In both languages, the alternation is available when a direct object is relativized, but not when an indirect object is relativized while a direct object remains in the same clause.

The last parallel, in particular, suggests a similar relationship between *wh*-displacement and the availability of the alternation in the two languages.

These striking parallels provide strong empirical evidence that the Japanese *-ga/-no* alternation and French stylistic inversion are syntactically the same phenomenon, despite their different morphological manifestations.

(e) There is, however, an apparent difference between the two alternations in long-distance *wh*-displacement of an object.

In French, stylistic inversion is available in the most deeply embedded clause involved in the displacement. In Japanese, by contrast, the *-ga/-no* alternation is not available on the subject of that clause, but it is available on the matrix subject. This difference must be acknowledged. Nevertheless, given the striking parallels observed in other environments, we may tentatively maintain the hypothesis that the two alternations are syntactically the same phenomenon, and ask how this difference might be accounted for.

4. (a) By the locality of head movement and the cyclicity of derivation, V-to-T movement before the PF–LF split requires, and is constituted by, both V-to-AgrO movement and AgrO-to-T movement before the split. Thus, it suffices to show that checking the accusative Case of the object before the PF–LF split implies each of these two movements before the split.

(i) Necessity of V-to-AgrO movement.

We first prove that checking the accusative Case of the object before the PF–LF split requires V-to-AgrO movement before the split. We prove this implication by contraposition. Suppose that V-to-AgrO movement does not take place before the PF–LF split.

We first establish the following lemma.

Lemma 1. Under the assumption that V-to-AgrO movement has not taken place, for the object to move to Spec-AgrO, the subject must have been extracted from VP beforehand.

> *Proof.* Suppose that the object moves from its θ-position to Spec-AgrO while the subject remains within VP. Both the subject's θ-position within VP and Spec-AgrO are A-positions, and the latter c-commands the former. Moreover, in the absence of V-to-AgrO movement, these two positions are not equidistant from the object's θ-position. Thus, the subject's θ-position intervenes between the object's θ-position and Spec-AgrO. Since phrasal movement is restricted to substitution into specifier positions, the assumption of Relativized Minimality prevents the object from moving to Spec-AgrO. Therefore, the subject must have been extracted from VP before the object moves to Spec-AgrO. □

By the cyclicity of derivation, if the object moves to Spec-AgrO before the PF–LF split, the subject cannot have been extracted from VP beforehand. Thus, the necessary condition established in Lemma 1 cannot be satisfied. It follows that, without V-to-AgrO movement, the object cannot move to Spec-AgrO before the PF–LF split. Consequently, its accusative Case cannot be checked before the split.

Therefore, checking the accusative Case of the object before the PF–LF split requires V-to-AgrO movement before the split.

(ii) Necessity of AgrO-to-T movement.

We next prove that checking the accusative Case of the object before the PF–LF split requires AgrO-to-T movement before the split. We prove this implication by contraposition. Suppose that AgrO-to-T movement does not take place before the PF–LF split.

Assume, for contradiction, that the accusative Case of the object is checked before the split. By the Case-checking requirement, the object must have moved

to Spec-AgrO before the split. By the cyclicity of derivation, the subject is still within VP when the object undergoes this movement.

We now establish the following lemma.

Lemma 2. Under the assumption that AgrO-to-T movement has not taken place, for the subject to move to Spec-AgrS after the object has moved to Spec-AgrO, the subject must have been extracted from VP beforehand.

> *Proof.* Suppose that the subject subsequently moves from its θ-position within VP to Spec-AgrS. Both Spec-AgrO and Spec-AgrS are A-positions, and the latter c-commands the former. In the absence of AgrO-to-T movement, these two positions are not equidistant from the subject's θ-position. Thus, Spec-AgrO, occupied by the object, intervenes between the subject's θ-position and Spec-AgrS. Since phrasal movement is restricted to substitution into specifier positions, the assumption of Relativized Minimality prevents the subject from moving to Spec-AgrS. Therefore, for the subject to move to Spec-AgrS after the object has moved to Spec-AgrO, the subject must have been extracted from VP beforehand. □

By Lemma 2, in the absence of AgrO-to-T movement, the subject cannot move to Spec-AgrS after the object has moved to Spec-AgrO. Consequently, the nominative Case of the subject cannot be checked. The derivation therefore fails to converge, contrary to our assumption that we are considering a convergent derivation of an active transitive clause.

Thus, in a convergent derivation, checking the accusative Case of the object before the PF–LF split requires AgrO-to-T movement before the split.

From (i) and (ii), checking the accusative Case of the object before the PF–LF split requires both V-to-AgrO movement and AgrO-to-T movement before the split. By the cyclicity of derivation, these movements constitute V-to-T movement.

Therefore, checking the accusative Case of the object before the PF–LF split requires the main verb to undergo head movement to T in the same clause before the split. □

(b) We show that if the accusative Case of the object is checked while the subject remains within VP, the subject must pass through Spec-T on its way to Spec-AgrS. We prove the statement by contraposition.

Suppose that the subject does not pass through Spec-T on its way to Spec-AgrS.

Assume, for contradiction, that there exists a convergent derivation in which the accusative Case of the object is checked while the subject remains within VP.

In such a derivation, the object occupies Spec-AgrO while the subject remains in its θ-position within VP. Both Spec-AgrO and Spec-AgrS are A-positions, and the latter c-commands the former. Since T intervenes between AgrO and AgrS, head movement cannot render these two specifier positions equidistant from the subject's θ-position.

Since the subject does not pass through Spec-T by assumption, and since all phrasal movement is restricted to substitution into specifier positions, the assumption of Relativized Minimality prevents the subject from moving from its θ-position to Spec-AgrS.

Consequently, the nominative Case of the subject cannot be checked, and the derivation fails to converge. This contradicts the assumption that the derivation converges.

Therefore, if the subject does not pass through Spec-T on its way to Spec-AgrS, there is no convergent derivation in which the accusative Case of the object is checked while the subject remains within VP.

This establishes the contrapositive. Hence, if the accusative Case of the object is checked while the subject remains within VP in a convergent derivation, the subject must pass through Spec-T on its way to Spec-AgrS. □

(c) Recall that Spec-T is assumed to be unavailable after the PF–LF split. Suppose further that Spec-T is unavailable even before the split. Thus, Spec-T is unavailable throughout the derivation.

Under these assumptions, both questions have negative answers, as shown by the following two propositions.

Proposition 1. The accusative Case of the object of a verb in an active transitive clause cannot be checked before the PF–LF split.

> *Proof.* Suppose, for contradiction, that the accusative Case of the object is checked before the PF–LF split. By the cyclicity of derivation, the subject is still within VP when the accusative Case of the object is checked. By the result of (b), the subject must subsequently pass through Spec-T on its way to Spec-AgrS. However, Spec-T is unavailable both before and after the PF–LF split, by our assumptions. Thus, the subject cannot pass through Spec-T at any stage of the derivation. This is a contradiction. Therefore, the accusative Case of the object cannot be checked before the PF–LF split. □

Proposition 2. In an active transitive expletive construction, the object cannot bear accusative Case.

> *Proof.* Suppose, for contradiction, that there exists a convergent derivation of an active transitive expletive construction in which the object bears accusative Case. In such a construction, Spec-AgrS is occupied by the expletive. Since Spec-T is unavailable throughout the derivation, the thematic subject remains within VP. Since the derivation converges, the accusative Case of the object must eventually be checked. By the result of (b), however, checking the accusative Case of the object while the subject remains within VP requires the subject to pass through Spec-T on its way to Spec-AgrS. This is impossible, since Spec-T is unavailable both before and after the PF–LF split. Thus, the accusative Case of the object cannot be checked, contrary to the assumption that the derivation converges. Therefore, in an active transitive expletive construction, the object cannot bear accusative Case. □

Thus, both questions have negative answers. Under the assumption that Spec-T is unavailable even before the PF–LF split, the accusative Case of the object of a verb in an active transitive clause cannot be checked before the split. Furthermore, in active transitive expletive constructions, the object cannot bear accusative Case at all.

A note on Bures's generalization

Bures's generalization states that a language allows the accusative Case of a non-pronominal object of a transitive verb to be checked before the PF–LF split if and only if a object can bear accusative Case in a transitive expletive construction in that language.
The preceding arguments outline an approach to accounting for this generalization in terms of the availability of Spec-T. In particular, the availability of Spec-T is proposed as a condition underlying both properties.

(d) We prove that, for French stylistic inversion to be possible in the relevant active transitive constructions, the direct object must undergo *wh*-displacement. We prove this statement by contraposition.

Suppose that the direct object does not undergo *wh*-displacement.

By assumption, the subject remains within VP at the PF–LF split. By the cyclicity of derivation, whether the accusative Case of the object is checked before or after the split, it must be checked while the subject remains within VP.

By the result of (b), checking the accusative Case of the object under this condition requires the subject to subsequently pass through Spec-T on its way to Spec-AgrS.

However, by assumption, Spec-T is unavailable in French both before and after the PF–LF split. Thus, the subject cannot pass through Spec-T at any stage of the derivation.

It follows that the accusative Case of the direct object cannot be checked, and hence the derivation cannot converge.

This establishes the contrapositive. Therefore, for French stylistic inversion to be possible in the relevant active transitive constructions, the direct object must undergo *wh*-displacement. □

(e) We prove that, for the Japanese *-ga/-no* alternation to be possible in the relevant active transitive constructions, the direct object marked with *-o* must undergo *wh*-displacement. We prove this statement by contraposition.

Suppose that the direct object marked with *-o* does not undergo *wh*-displacement.

By assumption, the subject remains within VP at the PF–LF split. By the cyclicity of derivation, whether the accusative Case of the object is checked before or after the split, it must be checked while the subject remains within VP.

By the result of (b), checking the accusative Case of the object under this condition requires the subject to subsequently pass through Spec-T on its way to Spec-AgrS.

However, by assumption, Spec-T is unavailable in Japanese both before and after the PF–LF split. Thus, the subject cannot pass through Spec-T at any stage of the derivation.

It follows that the accusative Case of the direct object marked with *-o* cannot be checked, and hence the derivation cannot converge.

This establishes the contrapositive. Therefore, for the Japanese *-ga/-no* alternation to be possible in the relevant active transitive constructions, the direct object marked with *-o* must undergo *wh*-displacement. □

An unexpected theorem

Notice that the assumption that the direct object does not undergo *wh*-displacement, introduced in both (d) and (e) for the purpose of proving the contrapositive, plays no role in either proof.

In fact, the proofs establish something stronger than what we originally intended to show. From the result of (b), together with the assumptions that the subject remains within VP at the PF–LF split and that Spec-T is unavailable throughout the derivation, we can derive the impossibility of accusative Case checking without making any assumption about whether the direct object undergoes *wh*-displacement.

This means that, within the fragment of the theory considered here, the impossibility of accusative Case checking is a theorem under the stated assumptions. Consequently, merely allowing the direct object to undergo *wh*-displacement, without changing those assumptions, cannot alter this result.

This is an unexpected and nontrivial consequence of the theory, which becomes apparent through our attempt to prove the original statements by contraposition. It is also empirically problematic: French stylistic inversion and the Japanese *-ga/-no* alternation are possible in certain environments involving the *wh*-displacement of a direct object, contrary to what the theory predicts under these assumptions.

Problems of this kind, revealed through proof-writing, indicate that something must be wrong with the theory as a deductive system, at least insofar as it is intended to account for the observed contrasts. We must therefore reconsider some of the assumptions underlying these deductions.

On the role of proof-writing

The empirical generalization we seek to capture concerns the necessity of *wh*-extraction: for French stylistic inversion or the Japanese *-ga/-no* alternation to be possible in an active transitive clause, the direct object must undergo *wh*-extraction. This generalization does not, however, imply that *wh*-extraction of the direct object can rescue a derivation that would otherwise fail to converge.

Proving the necessity of applying a particular operation is different from constructing a successful derivation involving that operation. The latter establishes that one possible derivation exists, but does not establish that the application of the operation is necessary. This distinction is especially important in generative syntax, where additional operations or derivational options may be introduced when difficulties arise. An indirect proof allows us to investigate necessity relative to a fixed set of theoretical assumptions, without having to construct a successful derivation.

This distinction also suggests a criticism of Watanabe's (1996) argument. In particular, in Section 4.2 and the subsequent discussion,

Watanabe attempts not only to account for the necessity of direct-object *wh*-extraction, but also to show how such extraction can make the alternations possible. To this end, he introduces substantial extensions to the theoretical system. In particular, he allows a new type of phrasal movement, adjunction to AgrOP, and adopts the hypothesis that Case checking can take place in an adjoined position, rather than only in a Spec–head configuration. This extension of the structural conditions on Case checking also raises further questions about agreement, which require additional assumptions.

However, these additional assumptions are not required merely to formulate and investigate the necessity of *wh*-extraction under the original theoretical assumptions. As our proofs illustrate, attempting to establish this necessity by contraposition can already reveal important consequences of those assumptions, including the unexpected theorem identified above. This theorem tells us that the original system cannot account for the observed contrasts as it stands. Introducing adjunction may be one possible repair, but it is an extension of the system, not a consequence that follows logically from the original assumptions. Another possibility would be to modify the theory of minimality itself.

More generally, I wonder whether the emphasis on constructing successful derivations in generative syntax sometimes comes at the expense of investigating the logical consequences of a theory and preserving the precise content of the empirical generalizations it is intended to capture. The problem is not the construction of successful derivations itself. Rather, when difficulties can be overcome by introducing new operations and assumptions, we may lose sight of the distinction between what the observations actually require, what follows from the existing theory, and what is introduced as a possible repair.

This case illustrates how proof-writing can help us discover unexpected consequences of a deductive system before attempting to repair it. By distinguishing the discovery of an empirically undesirable consequence of a theory from the proposal of a possible repair, we can better understand both what the theory actually entails and what is involved in extending or revising it.